

AI & IOT INTEGRATED WIND TREE FOR OPTIMIZED MICRO WIND ENERGY

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Outline



- ☐ Abstract
- ☐ Introduction
- ☐ Literature Review
- ☐ Motivation
- ☐ Objective
- ☐ Existing Method
- ☐ Proposed Method
- ☐ Hardware & Software Involved
- ☐ Advantages And Application
- ☐ System Architecture/Proposed Work Block Diagram
- ☐ UML Diagrams (CSE & Allied)/DFD CSE & Allied / Prototype (Others)
- ☐ Implementation(part Of The Code)
- ☐ Testing/Simulation
- ☐ Results And Discussions
- ☐ Conclusions And Future Scope
- ☐ References

Final Review



ABSTRACT

- This project explores an innovative, nature-inspired approach to renewable energy generation by developing a wind tree system utilizing aero leaves.
- Designed to mimic the structure and aesthetics of a natural tree, this setup integrates small, lightweight aero-leaf turbines in place of leaves to capture wind from all directions.
- With the integration of “Internet of Things technology and AI-based monitoring”, the system ensures efficient energy tracking, fault detection, and real-time performance optimization.
- This model emphasizes affordability and accessibility, promoting clean energy adoption at the grassroots level.



INTRODUCTION

- Wind Tree technology is a nature-inspired solution that mimics the form and function of natural trees to harness wind energy efficiently using compact vertical axis turbines called aeroleaves.
- Each one acts as a miniature wind turbine that can generate electricity even from low speed winds in all directions, making it suitable for areas where traditional turbines are inefficient.
- Integrating Artificial Intelligence (AI) and Internet of Things (IoT) enhances real-time monitoring, predictive maintenance, and adaptive energy optimization in the system.
- This Project explores how combining smart systems and renewable energy principles can lead to a scalable and eco-friendly power generation solution.



LITERATURE SURVEY

REF. NO	TITLE	TECHNIQUE USED	PROS	CONS
1	Fabrication and performance analysis of the aero-leaf savonius wind turbine tree	Savonius rotor design, performance testing	Works efficiently, urban-friendly	Mechanical imbalance, require external battery
2	Nature-inspired designs in wind energy	Bio-mimicry analysis, computational simulation	Low-noise, useful for urban architecture	May increase fabrication complexity and cost
3	Small wind turbines and their potential for internet of things applications	Design & Evaluation of Small Wind Turbines	Sustainable Power Source for IOT	Wind Availability Constraints Low Power Output
4	Energy prediction of wind turbine using iot	Sensor Integration with IoT	Real-time Monitoring, Energy Forecasting	Limited Sensor Accuracy, Model Limitations
5	Optimizing Energy Flow in a Wind-PV-Battery Microgrid Using AI Techniques	Microgrid, Energy Management, Swarm Intelligence, Wind Power, Photovoltaics	Optimized energy flow means reduced dependency on expensive fossil-fuel backups and lower operational costs	Installing wind turbines, solar panels, battery systems, and sensors, along with AI infrastructure, requires a significant upfront investment.



LITERATURE SURVEY

REF. NO	TITLE	TECHNIQUE USED	PROS	CONS
6	A comprehensive IoT cloud-based wind station ready for real-time measurements and artificial intelligence integration	Weather station Wind measurements IoT wind station Cloud-base weather monitoring	Immediate awareness of wind conditions and turbine performance	Requires investment in IoT hardware, cloud services, and AI development.
7	Review of Artificial Intelligence-Based Design Optimization of Wind Power Systems	Evolutionary Algorithms (GA, PSO), Hybrid AI Approaches	Comprehensive coverage of optimization areas	Limited discussion on real-time deployment
8	AI-Enabled Metaheuristic Optimization for Predictive Management of Renewable Energy Production in Smart Grids	Hybrid Metaheuristics (LSTM-RL, RL-SA, CNN-PSO)	Suitable for load balancing and rural microgrids	May require significant training data
9	Internet of Things (IOT) Based Machine Learning Techniques for Wind Energy Harvesting	IoT Sensors, Wireless Sensor Networks (WSNs)	Predictive Maintenance, Energy Output Optimization	High Initial Cost, Data Security Risks
10	Small Wind Electric System Energy Saver	Permanent Magnet Generators (PMG)	Energy Independence, Easy Installation	Noise & Vibration, Limited Output



MOTIVATION

- Most wind turbines require large open areas and consistent wind flow, making them unsuitable for cities, roof tops, or rural households with limited land.
- Inspired by natural leaf structures, the concept of a "Wind Tree" using aero-leaves offers a new path that is both functional and aesthetic.
- There is limited practical implementation data on nature-inspired vertical axis turbines like the aero-leaf design making this an opportunity to contribute real-world testing and improvement.
- The project aligns with global efforts to promote green, decentralized, and noise-free energy solutions that are safe for humans, animals, and the environment.



OBJECTIVE

- The Main motive of this project is to design and fabricate a low-cost Wind Tree prototype that harnesses wind energy using multiple aero-leaves arranged in a tree-like structure.
- To integrate IoT sensors for real-time monitoring of wind speed, turbine RPM, and energy generation statistics.
- To apply basic AI or Machine Learning models to predict energy output patterns and optimize blade performance under varying wind conditions.



EXISTING METHOD

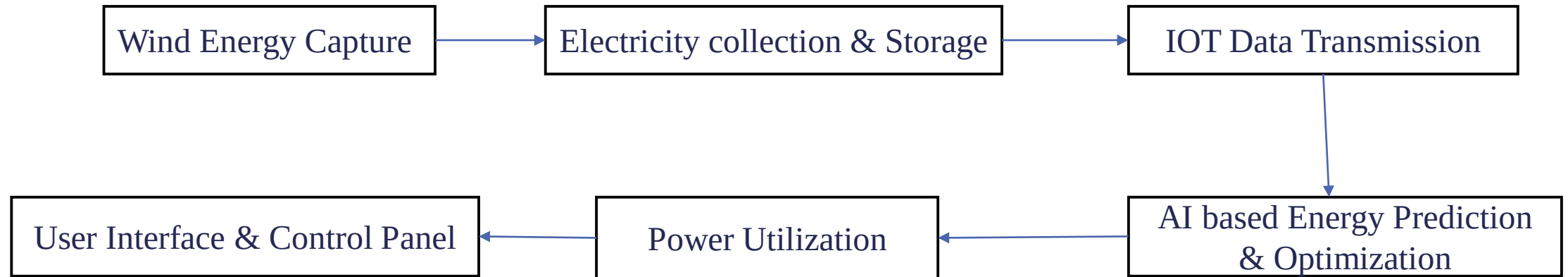
- Earlier introduced project include a wind turbine model shaped like a tree, using aéroleaf shaped Savonius rotors as the main energy-generating components.
- Each aero-leaf works as a small vertical-axis wind turbine (VAWT), capturing wind from all directions.
- The design focuses on low wind speed environments, making it ideal for urban and semi-urban use.
- The rotors are connected to small DC generators to produce electricity



PROPOSED METHOD

- Design and build a small-scale Wind Tree using aero-leaf-shaped Savonius rotors.
- Using recycled plastic, DC motors to keep minimal or zero cost.
- Integrating IoT sensors to monitor wind speed, power output, and rotor performance in real time.
- Develop a machine learning model to predict energy generation based on historical weather and Wind fan data.

BLOCK DIAGRAM





EXPLANATION OF PROPOSED WORK

What is the use?

- ✓ This project focuses on designing a nature-inspired wind energy harvesting system called a Wind Tree, helps reduce dependence on fossil fuels by using naturally available wind.
- ✓ Unlike traditional wind turbines, this system is compact, silent, and suitable for urban and rural areas where space and aesthetics are concerns.
- ✓ The system integrates Artificial Intelligence (AI) for efficient wind prediction and system optimization, while IoT (Internet of Things) is used for real-time monitoring of power generation, turbine health, and environmental data.



EXPLANATION OF PROPOSED WORK

How it is Going to be used in your Project?

- ✓ Assemble a small wind tree with aeroleaves, ESP32, sensors (wind, power, vibration, temp), and actuators for control.
- ✓ Connect nodes via Wi-Fi/LoRa/NB-IoT; send data using MQTT/HTTP; create a live monitoring dashboard.
- ✓ Gather wind and power data, then train ML models for forecasting, power prediction, anomaly detection, and control optimization.
- ✓ Run lightweight AI on the microcontroller for real-time decisions, while heavy training/analytics run in the cloud.
- ✓ Apply closed-loop AI control, measure improvements in energy output and reliability, and refine the system through iterations.



EXPLANATION OF PROPOSED WORK

Why You Choose and explain the advantages ?

- ✓ With rising pollution and climate change, there's a strong need to shift from fossil fuels to ecofriendly energy solutions like wind power.
- ✓ Unlike big wind turbines, this project works even with low or irregular wind flow, which is ideal for cities and villages.
- ✓ The idea of using a "wind tree" with aero-leaves is unique, visually appealing, and blends well with the environment.
- ✓ This project can be built using low-cost, reused items, making it affordable for students and communities.



HARDWARE & SOFTWARE INVOLVED

HARDWARE:

- DC Micro Generators
- Battery
- Microcontroller
- Charge Controller
- Anemometer
- RPM Sensor
- Voltage & Current Sensors
- Wi-Fi/Bluetooth Module
- Mounting Tree Structure

SOFTWARE:

- ✓ Arduino IDE

For programming microcontrollers and reading sensor data.

- ✓ ThingSpeak

Cloud-based platform for real-time IoT data visualization.

- ✓ Python / MATLAB

Used for data analysis, predictions, and AI-based forecasting.

- ✓ MicroPython

For coding the microcontroller logic.



APPLICATION & ADVANTAGES

APPLICATION:

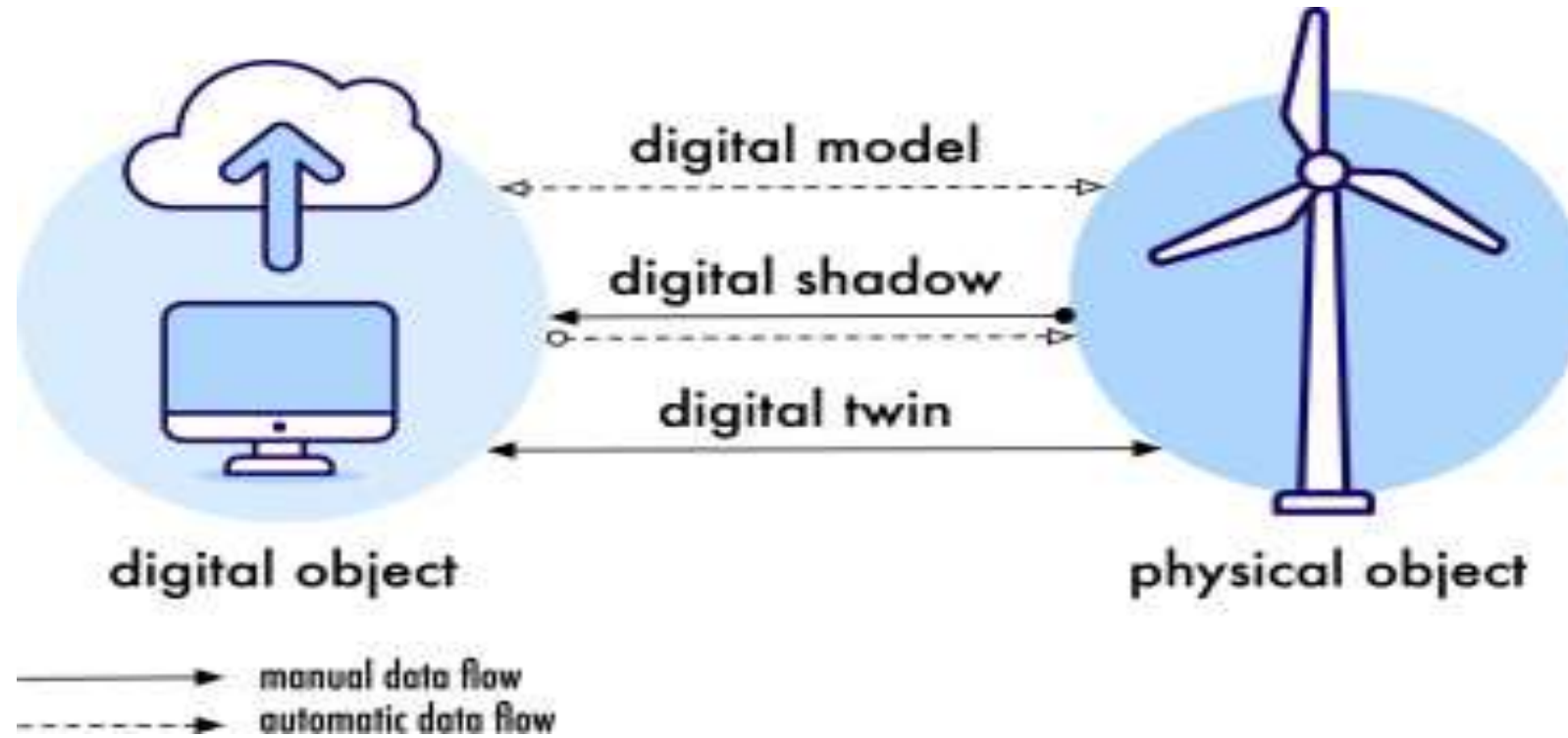
- This project supports sustainable infrastructure by powering streetlights, CCTV cameras, and IoT devices.
- It can provide basic electricity in remote villages or forest surveillance posts.
- It can acts as a natural energy backup for LED bulbs, fans, mobile charging, etc.
- It can be used in Environmental Monitoring System.



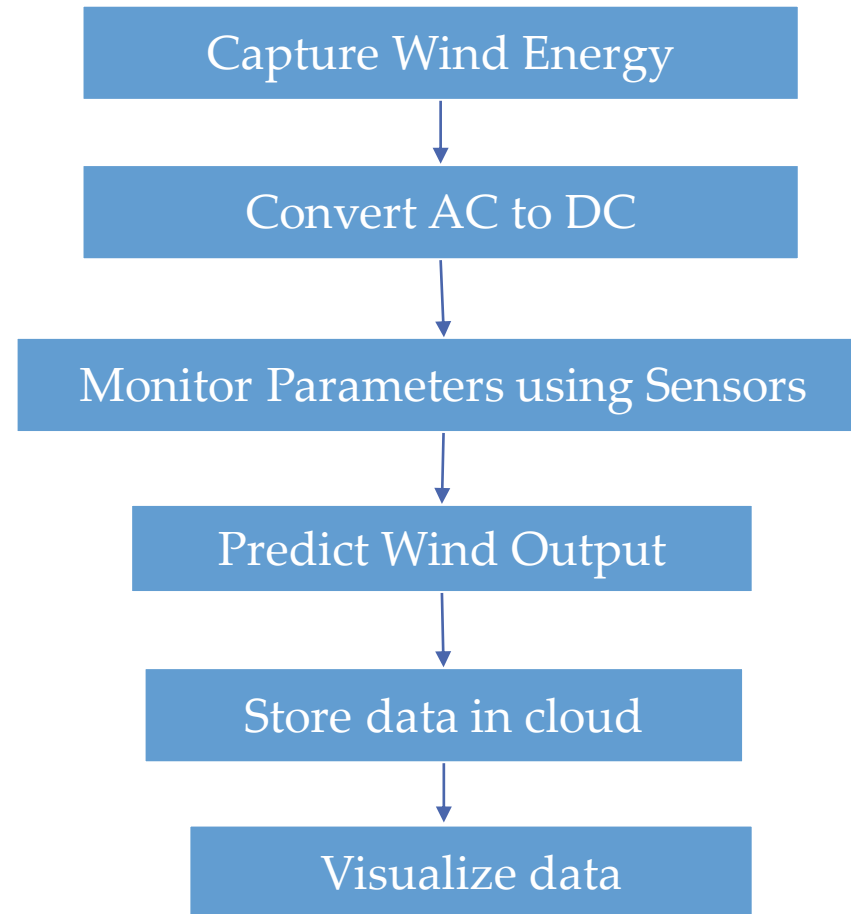
ADVANTAGES:

- It works silently unlike traditional horizontal turbines.
- We can increase the number of leaves (turbines) easily as per energy needs.
- Real-time power prediction, fault detection, and performance tracking from your phone or laptop.
- It blends well in public places without disturbing the visual appeal.

SYSTEM ARCHITECTURE



UML DIAGRAM



PROTOTYPE

- ✓ It's made using PVC pipes arranged like branches, and at the ends of these branches are mini turbine-like components probably made from small fans that act as aeroleaves or small vertical-axis wind turbines. These aeroleaves rotate with the wind to produce small amounts of electricity.
- ✓ At the base, there's a light or LED indicator, possibly showing that power is being generated or stored.

The setup seems to be a student project or prototype for a wind energy system that mimics the natural form of a tree, combining aesthetics with sustainability.





IMPLEMENTATION

- Connects to Wi-Fi using your Wi-Fi name and password.
- Prints dots (...) while trying to connect.
- Shows “Connected!” when Wi-Fi is successful.
- Connects to Firebase for storing or sending data online.
- Starts the INA219 sensor (used to measure voltage, current, and power).
- If the sensor is not found, it shows an error message.
- In the loop, it:
 - Reads Voltage
 - Read Current
- Prints all readings

```
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
Serial.print("Connecting to Wi-Fi");
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}
Serial.println(" Connected!");
Firebase.begin(FIREBASE_HOST, FIREBASE_AUTH);
Firebase.reconnectWiFi(true);
if (!ina219.begin()) {
  Serial.println("Failed to find INA219 chip");
  while (1) { delay(10); }
}
}

void loop() {
  busVoltage = ina219.getBusVoltage_V();
  current_mA = ina219.getCurrent_mA();
  power_mW = ina219.getPower_mW();
  Serial.print("Voltage: "); Serial.print(busVoltage); Serial.println("
V");
  Serial.print("Current: "); Serial.print(current_mA);
  Serial.println(" mA");
  Serial.print("Power: "); Serial.print(power_mW); Serial.println("
mW")
}
```



TESTING

STEPS:

- Upload the code and open Serial Monitor.
- Ensure your Firebase host URL and auth key are correct.
- If connected, no error will appear after `Firebase.begin()`.
- Will Blow air on your turbine (aeroleaves).
- Watch the current and power readings increase.
- Run the system for a few minutes.
- By checking with LED.

RESULTS



- During the testing and simulation phase, the system is connected to Wi-Fi and Firebase to verify proper communication.
- Once powered on, the LED bulb glows to indicate that the circuit is active and electricity is being generated by the mini wind turbine setup.
- The sensor continuously measures the voltage, current, and power produced by the small turbines. These readings are displayed in real time on the Serial Monitor, allowing you to observe changes in output as wind speed varies.
- If connected to Firebase, the same data can also be viewed remotely on the database dashboard, confirming successful data transmission.
- This simulation helps ensure that the wind tree prototype, sensors, and connectivity all function correctly before full deployment.



CONCLUSION & FUTURE SCOPE

- The glowing LED indicates real-time power generation, and with Firebase integration, data can be monitored remotely. This project proves that even small-scale, eco-friendly designs can contribute to sustainable energy solutions.
- In the future, the wind tree can be expanded by integrating AI-based data analysis to predict energy generation based on wind patterns.
- The system can be upgraded with IoT dashboards for live monitoring through mobile apps. Larger models can be installed in urban parks, campuses, and rural areas to supply clean power for streetlights or small devices.



REFERENCES

1. Patidar, H.; Shende, V.; Baredar, P.; Soni, A. Comparative study of offshore wind energy potential assessment using different Weibull parameters estimation methods. *Environ. Sci. Pollut. Res.* 2022, 29, 46341–46356. [CrossRef].
2. Bp Statistical Review of World Energy. 2022. Available online: <https://www.bp.com/en/global/corporate/energy-economics/statistical-review-of-worldenergy.html> (accessed on 15 January 2023).
3. Power Consumption to Triple by 20500 —McKinsey Report. Available online: <https://www.modernpowersystems.com/news/newspower-consumption-to-triple-by-2050-mckinsey-report9655086#> (accessed on 8 January 2023).
4. Amano, R.S. Review of Wind Turbine Research in 21st Century. *J. Energy Resour. Technol.* 2017, 139, 050801. [CrossRef].
5. Prabhukhot, P.R.; Prabhukhot, A.R. Computer Analysis of S822 Aerofoil Section for Blades of Small Wind Turbines at Low Wind Speed. *J. Sol. Energy Eng.* 2017, 139, 051008–51011. [CrossRef].
6. Global Energy Review 2020 by IEA. Available online: <https://www.iea.org/reports/global-energy-review-2020> (accessed on 15 January 2023).
7. Using numerical and metaheuristic optimization methods for different Indian terrains. *Environ. Sci. Pollut. Res.* 2023, 30, 30874–30891. [CrossRef].



REFERENCES

8. Mishnaevsky, L.; Hasager, C.B.; Bak, C.; Tilg, A.-M.; Bech, J.I.; Rad, S.D.; Fæster, S. Leading edge erosion of wind turbine blades: Understanding, prevention and protection. *Renew. Energy* 2021, 169, 953–969. [CrossRef].
9. Chandrakant, S.; Panchal, H.; Sadasivuni, K.K. Numerical simulation of flow-through heat exchanger having helical flow passage using high order accurate solution dependent weighted least square based gradient calculations. *Energy Sources Part A Recover. Util. Environ. Eff.* 2021, 1–26. [CrossRef]
10. Tian, W.; Song, B.; VanZwieten, J.H.; Pyakurel, P. Computational Fluid Dynamics Prediction of a Modified Savonius Wind Turbine with Novel Blade Shapes. *Energies* 2015, 8, 7915–7929. [CrossRef]
11. Patel, A.; Soni, A.; Baredar, P.; Malviya, R. Analysis of temperature distribution over pipe surfaces of air-based cavity linear receiver for cross-linear concentration solar power system. *Environ. Sci. Pollut. Res.* 2022, 1–19. [CrossRef] [PubMed]
12. Blackwell, B.; Sheldahl, R.; Feltz, L. Wind Tunnel Performance Data for Two-and-Three-Bucket Savonius Rotors; US Sandia Laboratories Report SAND76-0131; Sandia Laboratories: Springfield, VA, USA, 1977.



THANK YOU